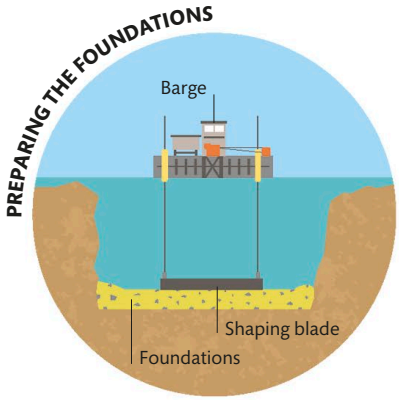
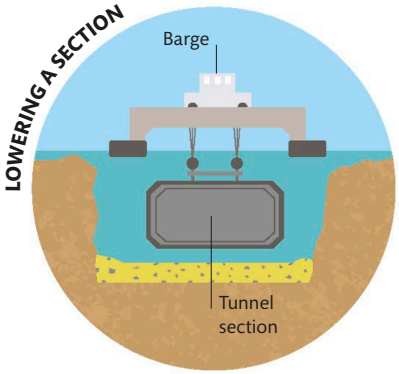
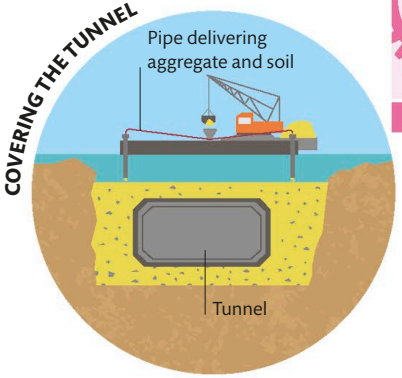


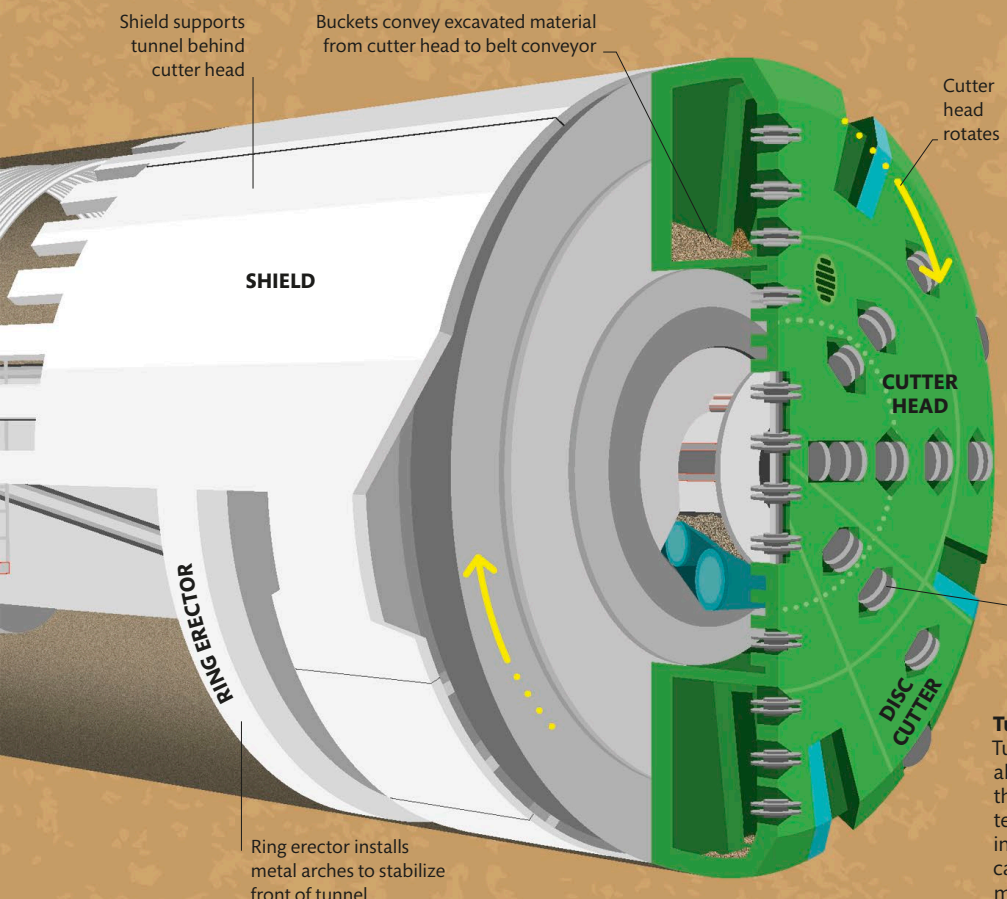




2 The bottom of the trench is prepared by laying a foundation of aggregate and sand. The foundation is smoothed with a shaping blade to ensure a level base for the tunnel sections.

3 Prefabricated cast-concrete sections are floated to the installation site and lowered to the bottom. A hydraulic arm pulls each new section close up to the adjacent section to form a watertight seal.

4 Pipes running from a barge deliver more aggregate and soil to cover the completed tunnel. The top of the tunnel may also be covered by a layer of large stones to protect it from damage by ship anchors.



35 MILES (57 KM)

THE LENGTH OF THE **WORLD'S LONGEST RAIL TUNNEL**, THE TWIN-TUBED **GOTTHARD BASE TUNNEL** UNDER THE SWISS ALPS

Tunnel boring machine
Tunnel boring machines can bore through all types of earth, including hard rock. At the front is a circular plate with cutting teeth. As the plate rotates, the teeth slice into the rock, which falls onto a conveyor carrying the debris to the rear of the machine. As the machine moves forward, it lines the tunnel with concrete.